"domain of the large," where general relativity must be highly accurate; and the general relativistic laws were unequivocal—nothing can escape out of the critical circumference. Gravity holds everything in. Thus, there can be no "quantum mechanical tunneling" (whatever that was) to let radiation out; I was firmly convinced of it.

In 1964 and 1965 Wheeler and I wrote a technical book, together with Kent Harrison and Masami Wakano, about cold, dead stars and stellar implosion. I was shocked when Wheeler insisted on including in the last chapter his speculation that radiation might tunnel its way out of the hole and escape into interstellar space. In a last-minute struggle to convince Wheeler to delete his speculation from the book, I called on David Sharp, one of Wheeler's postdocs, for help. David and I argued vigorously with Wheeler in a three-way telephone call, and Wheeler finally capitulated.

Wheeler was right; David and I were wrong. Ten years later, Zel'-dovich and Stephen Hawking would use a newly developed partial marriage of general relativity and quantum mechanics to prove, mathematically, that radiation *can* tunnel its way out of a black hole—though very, very slowly (Chapter 12). In other words, black holes can evaporate, though they do it so slowly that a hole formed by the implosion of a star will require far longer than the age of our Universe to disappear.

The names that we give to things are important. The agents of movie stars, who change their clients' names from Norma Jean Baker to Marilyn Monroe and from Béla Blasko to Béla Lugosi, know this well. So do physicists. In the movie industry a name helps set the tone, the frame of mind with which the viewer regards the star—glamour for Marilyn Monroe, horror for Béla Lugosi. In physics a name helps set the frame of mind with which we view a physical concept. A good name will conjure up a mental image that emphasizes the concept's most important properties, and thereby it will help trigger, in a subconscious, intuitive sort of a way, good research. A bad name can produce mental blocks that hinder research.

Perhaps nothing was more influential in preventing physicists, between 1939 and 1958, from understanding the implosion of a star than the name they used for the critical circumference: "Schwarzschild singularity." The word "singularity" conjured up an image of a region where gravity becomes infinitely strong, causing the laws of physics as we know them to break down—an image that we now understand is correct for the object at the